
SRUL: Spatial Spherical Latents with Riemannian Flow Matching

Mohammadreza Tavasoli Naeini

Department of Electrical and Computer Engineering, University of Toronto
mohammadreza.tavasolinaeini@mail.utoronto.ca

Abstract

Autoencoders can learn latent spaces that are not Gaussian. When normalization is used, their features may concentrate near a fixed-radius shell; Transformer autoencoders with LayerNorm are one important example. Ignoring this geometry can make the generative prior move through latent regions that do not match the encoder output. We use *Spherical Riemannian Unified Latents* (SRUL) to examine this problem in a controlled setting. SRUL makes spatial latent tokens exactly spherical and models them with Riemannian Flow Matching. On CIFAR-10, replacing a Euclidean chord with a geodesic path improves FID from 67.44 to 63.84 while keeping the encoder and decoder unchanged. We also study tangent-noise scale and classifier-free guidance, and evaluate the same approach on PathMNIST and CelebA-64.

Attestation of Teamwork

Mohammadreza Tavasoli Naeini conducted the background study, designed and implemented the models, ran and analyzed the experiments, and prepared the report, presentation, and code.

1 Introduction

Latent generative models use an autoencoder to compress an image and a prior to generate new latent codes. The geometry of these codes is learned by the encoder and does not have to be Gaussian. In particular, normalization can make feature norms nearly constant, so the latent representation becomes close to a hypersphere. Transformer autoencoders with LayerNorm are one common example, while explicit feature normalization can produce the same structure in other architectures.

This geometry should not be ignored when the prior is designed. If the encoder produces fixed-radius latents but the prior uses ordinary Euclidean interpolation, the intermediate states can leave the region represented by the encoder. Even with normalized endpoints, a straight path forms a chord through the sphere rather than moving along its surface.

The same mismatch can be stated at the distribution level. Let $q_\phi(z)$ denote the aggregate distribution of encoded latents. For any prior $p(z)$,

$$\mathbb{E}_x[\text{KL}(q_\phi(z | x) \| p(z))] = I_q(X; Z) + \text{KL}(q_\phi(z) \| p(z)). \quad (1)$$

The first term is the image information retained by the latent. The second term is the mismatch between the encoder distribution and the chosen prior. When every token has fixed norm, a prior or transport path that introduces radial variation models states that the encoder does not normally produce.

We use SRUL as a controlled way to examine this issue. It makes the spherical latent geometry exact, then compares a chord-based prior with a geometry-consistent Riemannian prior while keeping the encoder and decoder fixed. We also study tangent-noise scale and classifier-free guidance, and evaluate the final method on CIFAR-10, PathMNIST, and CelebA-64.

ECE 1508: Deep Generative Models (Summer 2026)

2 Related Work and Positioning

Latent Diffusion Models (LDMs) move generation from pixels to a learned Euclidean latent grid, which reduces computation while preserving perceptually important information [1]. Unified Latents studies how encoder noise controls the amount of information passed through the latent representation, but its generative process remains Euclidean [5]. Sphere Encoder explicitly maps images to a sphere and trains the decoder to remain stable around encoded points, with the goal of direct sampling from an approximately uniform spherical distribution [4].

Flow Matching learns a time-dependent velocity field along a chosen probability path [2]. RJF observes that normalized representation encoders often produce nearly fixed-radius features and replaces Euclidean interpolation with Riemannian Flow Matching on the sphere; it also studies an additional Jacobi weighting [3]. SRUL builds on these directions but learns the representation and prior together as one pipeline. It uses a spatial spherical autoencoder, applies controlled tangent perturbations, and learns the nonuniform distribution of encoded tokens with an RFM prior. Unlike direct spherical sampling, the model does not assume that all points on the product of spheres are equally probable.

3 Method

3.1 Pipeline overview

Figure 1 shows the complete SRUL pipeline. Representation learning produces a grid of unit-norm tokens and trains the decoder to remain stable under tangent perturbations. RFM then learns the distribution of these encoded tokens using geodesic paths and tangent velocities. During sampling, spherical noise is transported with exponential-map steps and decoded into a new image.

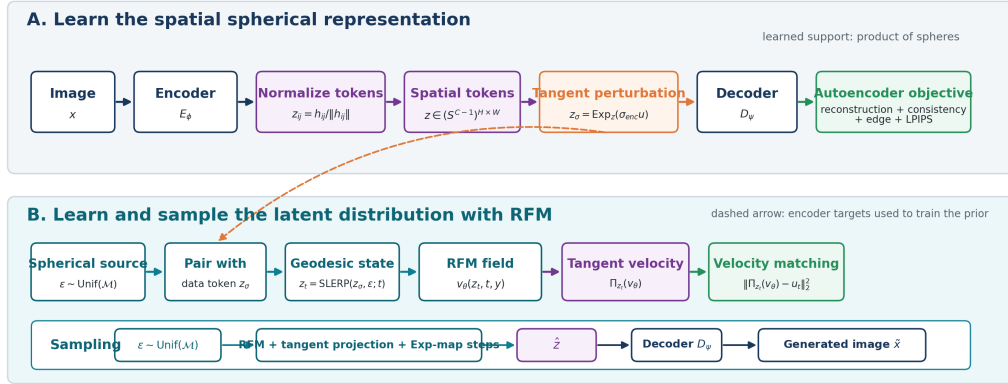


Figure 1: SRUL pipeline. The top stage learns spatial spherical tokens and a decoder robust to tangent perturbations. The lower stage trains and samples an RFM prior on the same product-of-spheres manifold. Class labels and classifier-free guidance are optional conditional inputs.

3.2 Spatial spherical autoencoder

For an image x , the encoder produces a spatial tensor

$$h = E_\phi(x) \in \mathbb{R}^{C \times H \times W}, \quad z_{:,i,j} = \frac{h_{:,i,j}}{\|h_{:,i,j}\|_2} \in S^{C-1}. \quad (2)$$

The full latent space is the product manifold $\mathcal{M} = (S^{C-1})^{HW}$. We use $C = 32$, a 4×4 token grid for 32×32 images, and an 8×8 grid for CelebA-64. The spatial layout allows different tokens to retain local image information, while normalization removes the radial degree of freedom.

The autoencoder is trained from clean and perturbed latents:

$$\begin{aligned}\mathcal{L}_{\text{AE}} = & \lambda_c \|D_\psi(z) - x\|_1 + \lambda_n \|D_\psi(z_\sigma) - x\|_1 + \lambda_e \mathcal{L}_{\text{edge}} \\ & + \lambda_{\text{lat}} [1 - \cos(z, E_\phi(D_\psi(z_\sigma)))] + \lambda_p \mathcal{L}_{\text{LPIPS}}.\end{aligned}\quad (3)$$

The clean and noisy reconstruction terms preserve image content, while latent consistency, edge loss, and LPIPS improve local stability and perceptual detail.

3.3 Tangent-noise control

For each token, Gaussian noise is projected onto the tangent space,

$$\xi \sim \mathcal{N}(0, I), \quad u = \Pi_z(\xi) = \xi - \langle \xi, z \rangle z, \quad (4)$$

and the perturbed token is

$$z_\sigma = \text{Exp}_z(\sigma_{\text{enc}} u). \quad (5)$$

Because $u \perp z$, the operation changes direction without changing radius. The scalar σ_{enc} therefore gives direct control over angular perturbation and retained image information.

3.4 Riemannian Flow Matching prior

The source is uniform on the same manifold, $\epsilon \sim \text{Unif}(\mathcal{M})$. For a data latent z_σ and source sample ϵ , the tokenwise geodesic is

$$z_t = \text{SLERP}(z_\sigma, \epsilon; t) = \frac{\sin((1-t)\Omega)}{\sin \Omega} z_\sigma + \frac{\sin(t\Omega)}{\sin \Omega} \epsilon, \quad \Omega = \arccos(z_\sigma^\top \epsilon). \quad (6)$$

Let $u_t = \partial z_t / \partial t$ be the target tangent velocity. We train

$$\mathcal{L}_{\text{RFM}} = \mathbb{E} \left[\left\| \Pi_{z_t}(v_\theta(z_t, t)) - u_t \right\|_2^2 \right]. \quad (7)$$

During sampling, each predicted velocity is projected and integrated with the exponential map,

$$z_{t-\Delta t} = \text{Exp}_{z_t}(-\Delta t \Pi_{z_t}(v_\theta)), \quad (8)$$

so every intermediate token remains on its sphere.

3.5 Conditional generation and classifier-free guidance

Conditioning is added without changing the spherical geometry. The field receives a label y , giving $v_\theta(z_t, t, y)$. During training, the label is replaced by a null label with probability 0.1. During sampling,

$$v_{\text{CFG}} = v_\theta(z_t, t, \emptyset) + s[v_\theta(z_t, t, y) - v_\theta(z_t, t, \emptyset)]. \quad (9)$$

The guided field is projected onto the tangent space before every exponential-map update. CFG therefore changes conditional sampling strength but not the manifold or probability path.

4 Experiments

Setup

CIFAR-10 is used for controlled comparisons [6]. PathMNIST tests transfer to texture-dominated histopathology [7], and CelebA-64 tests higher-resolution facial structure [8]. AdamW is used in both training stages, and final samples use 100 integration steps. We report FID and KID for distribution quality, feature precision and recall for fidelity and coverage, and token norm for geometry.

Table 1: Datasets and model configurations.

Dataset	Train	Resolution	Token grid	AE/prior epochs	Conditioning
CIFAR-10	50,000	32^2	4×4	60/120	class label
PathMNIST	89,996	32^2	4×4	50/80	tissue class
CelebA-64	30,000	64^2	8×8	40/60	smiling attribute

Does the transport path match the encoder geometry?

The main comparison keeps the spherical encoder, decoder, source distribution, latent size, prior capacity, and training budget fixed. Only the path changes. The baseline uses

$$z_t^{\text{chord}} = (1 - t)z_\sigma + t\epsilon. \quad (10)$$

Although both endpoints have unit norm, the chord enters the sphere. For nearly orthogonal endpoints, its midpoint has norm $\|(z_\sigma + \epsilon)/2\|_2 \approx 1/\sqrt{2} = 0.707$. The measured minimum mean norm is 0.705. SRUL instead follows SLERP and remains at norm 1.

Same spherical endpoints, different transport path

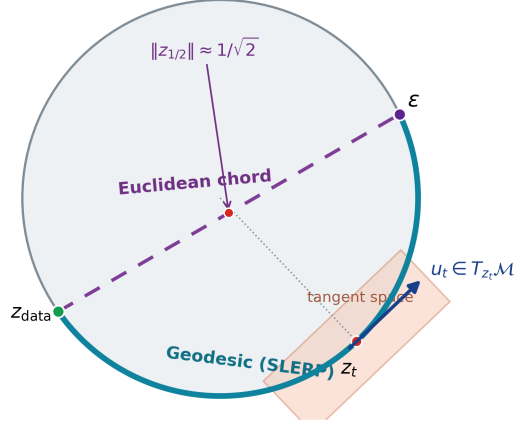


Figure 2: The chord visits interior states that the spherical encoder does not produce. SRUL follows the geodesic, and its velocity lies in the tangent space.

Table 2: CIFAR-10 controlled comparison with the same spherical encoder and decoder.

Method	FID↓	KID↓	Precision↑	Recall↑	Min. norm↑
Chord baseline	67.44	0.0740	0.840	0.085	0.705
SRUL	63.84	0.0667	0.857	0.091	1.000

The geodesic path improves all four image metrics and removes the radial excursion. This supports the central hypothesis: once the representation is spherical, its generative transport should remain on the manifold.

Control ablations

We train a separate conditional SRUL prior for each tangent-noise value. Increasing σ_{enc} removes more image information and worsens both reconstruction and generation; 0.05 is the best tested value. We then vary CFG while keeping the representation and RFM objective fixed. Stronger guidance improves FID and precision, but reduces recall.

Table 3: CIFAR-10 tangent-noise and classifier-free-guidance ablations.

σ_{enc}	Recon. FID↓	Gen. FID↓	Recall↑	CFG s	FID↓	Precision↑	Recall↑
0.05	17.53	48.71	0.090	1.0	62.44	0.854	0.083
0.15	19.56	51.61	0.090	1.5	54.83	0.865	0.071
0.30	27.85	58.16	0.073	2.0	49.39	0.884	0.059

Evaluation across image domains

We applied the same spatial spherical representation and RFM prior in three settings. CIFAR-10 provides the controlled geometry and sampling ablations, PathMNIST evaluates local histopathology

texture, and CelebA-64 tests a larger token grid on structured facial images. Table 4 summarizes the selected configuration and its quantitative results in each setting.

Table 4: Selected SRUL results in three evaluation settings. Reconstruction and generation metrics are computed against the real split of the corresponding dataset.

Dataset	Evaluation setting	Grid / selected con- trols	Recon. FID	Gen. FID	Precision	Recall
CIFAR-10	Natural-object geometry and sampling study	4×4 ; $\sigma_{\text{enc}} = 0.05$, $s = 2$	17.53	48.71	0.861	0.090
PathMNIST	Texture-dominated histopathology transfer	4×4 ; $\sigma_{\text{enc}} = 0.15$, $s = 2$	7.13	29.36	0.829	0.309
CelebA-64	64×64 facial-image scale-up	8×8 ; $\sigma_{\text{enc}} = 0.15$, $s = 2$	9.08	41.22	0.657	0.198

On CIFAR-10, the selected result follows the controlled chord-versus-geodesic comparison and the noise and guidance ablations. PathMNIST shows that the same 4×4 token design can model fine tissue texture. CelebA-64 increases the latent grid to 8×8 and preserves coherent facial structure at twice the image resolution.

5 Conclusion

SRUL learns spatial spherical tokens and models them with an RFM prior. With the encoder and decoder fixed, replacing the Euclidean chord by a geodesic improves CIFAR-10 FID from 67.44 to 63.84 and keeps every token on the manifold. The best tested noise scale is $\sigma_{\text{enc}} = 0.05$, and classifier-free guidance provides a separate fidelity–diversity control. PathMNIST and CelebA-64 show that the same design transfers to tissue texture and higher-resolution faces.

References

- [1] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer. High-resolution image synthesis with latent diffusion models. CVPR, 2022.
- [2] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le. Flow matching for generative modeling. ICLR, 2023.
- [3] A. Kumar and V. M. Patel. Learning on the Manifold: Unlocking Standard Diffusion Transformers with Representation Encoders. arXiv, 2026.
- [4] Z. Yue et al. Image Generation with a Sphere Encoder. arXiv, 2026.
- [5] J. Heek et al. Unified Latents: Controlling Information Flow in Latent Generative Models. arXiv, 2026.
- [6] A. Krizhevsky. Learning Multiple Layers of Features from Tiny Images. Technical report, 2009.
- [7] J. Yang, R. Shi, D. Wei, et al. MedMNIST v2: A Large-Scale Lightweight Benchmark for 2D and 3D Biomedical Image Classification. Scientific Data, 2023.
- [8] Z. Liu, P. Luo, X. Wang, and X. Tang. Deep Learning Face Attributes in the Wild. ICCV, 2015.